

CFD 2025-26 – Lab 09

Exercise 1

Let us consider the steady Navier-Stokes problem defined in the unit square $\Omega := (0, 1)^2$

$$\begin{cases} (\mathbf{u} \cdot \nabla) \mathbf{u} - \frac{1}{\text{Re}} \Delta \mathbf{u} + \nabla p = 0, & \text{in } \Omega, \\ \text{div } \mathbf{u} = 0, & \text{in } \Omega, \\ \mathbf{u} = 1\mathbf{i} + 0\mathbf{j}, & \text{on } \Gamma^{\text{up}} = \{0 \leq x \leq 1, y = 1\}, \\ \mathbf{u} = \mathbf{0}, & \text{on } \partial\Omega \setminus \Gamma^{\text{up}}. \end{cases}$$

- 1 Using the equal-order pair of spaces $\mathbb{P}^1/\mathbb{P}^1$, solve the problem using the Picard fixed-point iteration method. Each nonlinear iteration can be solved using a direct method (`param = {'ksp_type': 'preonly', 'pc_type': 'lu', 'pc_factor_mat_solver_type': 'mumps'}`). Solve the problem with $\text{Re} = 30$. Does the solver yield a proper solution?
- 2 Construct the nullspace of the monolithic matrix associated to the fully Dirichlet boundary conditions and pass it to the solver. Do you notice any difference?
- 3 The Brezzi-Pitkäranta stabilization of the Picard iteration reads as follows:

$$a(\mathbf{u}_k, \mathbf{v}) + c(\mathbf{u}_{k-1}, \mathbf{u}_k, \mathbf{v}) + b(\mathbf{v}, p_k) - b(\mathbf{u}_k, q) + \sum_{K \in \mathcal{T}_h} \delta_K (\nabla p_k, \nabla q)_K = 0$$

where $(\cdot, \cdot)_K$ denotes the L^2 inner product over the element K , and the stabilization parameter is

$$\delta_K = \delta h_K^2 \text{Re},$$

where $\delta = 0.01$ is a given parameter.¹ Implement this stabilization as an addition to the forms already introduced in the previous points and compare the results, both in terms of iterations needed to reach convergence, and the obtained pressure and velocity field.

- 4 Consider now different values of the Reynolds number: $\text{Re} = 3, 30, 300, 3000$. Comment on the differences in the results.

¹Notice that, for the problem at hand, $\nu = \frac{1}{\text{Re}}$.

- 5 The SUPG stabilization of the linearized problem can be written as follows:

$$\begin{aligned}
& a(\mathbf{u}_k, \mathbf{v}) + c(\mathbf{u}_{k-1}, \mathbf{u}_k, \mathbf{v}) + b(\mathbf{v}, p_k) - b(\mathbf{u}_k, q) \\
& + \sum_{K \in \mathcal{T}_h} \delta_K \left(-\frac{1}{\text{Re}} \Delta \mathbf{u}_k + (\mathbf{u}_{k-1} \cdot \nabla) \mathbf{u}_k + \nabla p_k, (\mathbf{u}_{k-1} \cdot \nabla) \mathbf{v} + \frac{1}{2} \text{div}(\mathbf{u}_{k-1}) \mathbf{v} + \nabla q \right)_K \\
& + \sum_{K \in \mathcal{T}_h} \delta_K (\text{div } \mathbf{u}_k, \text{div } \mathbf{v})_K \\
& = 0
\end{aligned}$$

where $(\cdot, \cdot)_K$ denotes the L^2 inner product over the element K . The stabilization parameter should depend on the local Reynolds number Re_K :¹

$$\delta_K = \begin{cases} \delta \frac{h_K}{\bar{u}_K}, & \text{for } \text{Re}_K \geq 1, \\ \delta h_K^2 \text{Re}_K, & \text{for } \text{Re}_K < 1, \end{cases}$$

where δ is a given scalar and

$$\text{Re}_K = \bar{u}_K h_K \text{Re}, \quad \bar{u}_K = \frac{1}{|K|} \int_K |\mathbf{u}_k| dx,$$

or, for brevity, $\delta_K = \delta \frac{h_K}{\bar{u}_K} \min\{1, \text{Re}_K\}$. Implement this stabilization, with $\delta = 0.01$ and with $\delta = 1$, and compare the results with those obtained at the previous points. Beware that $\bar{u}_K$ can reach very small values, thus the computation of δ_K should be carefully dealt with.

Exercise 2 - homework

We want to compare different choices for the diffusion term and the boundary conditions in the following problem over $\Omega = (0, 3) \times (0, 1)$:

$$\begin{cases} (\mathbf{u} \cdot \nabla) \mathbf{u} - \frac{1}{\text{Re}} \text{div } G(\mathbf{u}) + \nabla p = \mathbf{0}, & \text{in } \Omega, \\ \text{div } \mathbf{u} = 0, & \text{in } \Omega, \\ \mathbf{u} = 6y(1-y)\mathbf{i}, & \text{on } \Gamma_{\text{left}} := \{x=0, 0 \leq y \leq 1\}, \\ \frac{1}{\text{Re}} G(\mathbf{u}) \mathbf{n} - p \mathbf{n} = 0, & \text{on } \Gamma_{\text{right}} := \{x=3, 0 \leq y \leq 1\}, \\ \mathbf{u} = \mathbf{0}, & \text{on } \Gamma_{\text{bottom}} := \{0 \leq x \leq 3, y=0\}, \\ \star, & \text{on } \Gamma_{\text{top}} := \{0 \leq x \leq 3, y=1\}, \end{cases}$$

where the possible choices for the operator G and the boundary condition $\star$ are:

$$\begin{array}{ll} G(\mathbf{u}) = \nabla \mathbf{u} & \text{vs.} \quad G(\mathbf{u}) = \varepsilon(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T) \\ \star : \text{no-slip} & \text{vs.} \quad \star : \text{free slip} \\ \mathbf{u} = \mathbf{0} & \mathbf{u} \cdot \mathbf{n} = 0, \quad \left(\frac{1}{\text{Re}} G(\mathbf{u}) - p \mathbf{I} \right) \mathbf{n} \cdot \mathbf{t} = 0 \end{array}$$

- 1 For each combined choice of G and $\star$, derive the weak form of the problem.
- 2 Modify the code of Lab08 to solve the problem, and compare the results associated with the different choices. How would you model the flow between two infinite, rough planes?